NICHOLAS SOBCHAK

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Portfolio | LinkedIn | GitHub | U.S. Citizen, Eligible for U.S. Security Clearance

EXPERIENCE

Founding Software Developer – ZipTip Dec 2025 – Present
Remote / Westminster, MD

- Designed a serverless scan API (AWS Lambda, API Gateway, DynamoDB) with custom-domain routing, authentication, and ID-to-account redirect logic for a real-time tipping platform.
- Upgraded a large Angular codebase with shared linting tooling, typed API clients, and service-injection refactors for a better developer experience.
- Built AWS SAM CI/CD pipelines with OIDC-based auth, automated Route 53 DNS propagation after deploy, and rollback support for dev and prod environments.
- Implemented Open Graph link previews with bot/crawler detection and pre-signed S3 URLs for rich social shares of tip links.

SKILLS

JavaScript / TypeScript: Angular, React, React Native, HTML, CSS
Python: FastAPI, Django, Flask, pytest
Java: Spring Boot, JPA, Gradle, JUnit
AWS & Cloud: Lambda, SAM CLI, boto3, Cognito, CloudFormation, API Gateway, Docker
Database: PostgreSQL, DynamoDB, SQLite
Tools: Git/GitHub, GitHub Actions, Linux, ESLint, Prettier, C++17 (CMake)

TECHNICAL PROJECTS GitHub

On The Money – *TypeScript (React/Vite PWA); Java (Spring Boot), JPA, PostgreSQL; C++ Monte Carlo engine; Docker; Plaid*

- Integrated Plaid Link to let users securely connect bank accounts and automatically sync real transactions and balances through Plaid's API.
- Built a C++17 Monte Carlo engine running `10,000 simulations/request` into a Spring Boot REST API for retirement and portfolio growth projections.
- Pulled live stock quotes, market indices, and symbol search via Finnhub with a per-user watchlist; shipped the frontend as a React (Vite) PWA and PostgreSQL database on Vercel, with a Spring Boot API on Render. (see On The Money)

QuickQuill – *TypeScript (Angular); Java (Spring Boot), JPA, PostgreSQL; Docker; C++17;*

- Built a C++17 trie-based engine that serves lookups, spell-check suggestions, and autocomplete across a `1.28M-word` dictionary using thread-local SQLite connections and a thread-safe in-memory LRU cache.
- Integrated the native engine into a Spring Boot REST API via Java's Panama FFM (Foreign Function & Memory) API with a flat `extern "C"` ABI — native calls with no JNI or glue code.
- Shipped an Angular 21 frontend on Vercel with RxJS debounced search streams and ghost-text autocomplete; multi-stage Dockerized deployment of the C++ engine, Spring Boot, and PostgreSQL on Render. (see QuickQuill)

EDUCATION

University of Maryland, Baltimore County Aug 2025 – May 2027
Bachelor of Science in Computer Science
Relevant coursework: Neurosymbolic AI

Carroll Community College Sept 2023 – May 2025
Associate of Arts in Computer Science